

Response to Reviewers — Second Cut

Team 0 · A Load-Cell Cup Detection Module

Acknowledgements

The authors thank the reviewers for their careful evaluation. Line, section, and figure numbers refer to the revised manuscript. **Main changes:** the Introduction now states a research question (Section I), and Section III adds the module architecture (Fig. 1).

Summary of changes (observation–response–change matrix)

7 observations: 5 addressed, 1 partially addressed, 1 not addressed (justified).

Key	Status	Location in the revised paper
R1.1	[Addressed]	Section I, line 19, line 33
R1.2	[Addressed]	Whole manuscript
R2.1	[Addressed]	Lines 23–31
R3.1	[Addressed]	Fig. 1, Eq. (1), lines 40–51
R3.2	[Partially addressed]	Table I, lines 57–62
R4.1	[Addressed]	Lines 34–38
R4.2	[Not addressed (justified)]	Lines 64–69

Reviewer 1

Format, LaTeX and template compliance

R1.1 [Addressed]

Reviewer comment: Las fuentes están escritas a mano en formato autor–año.

Response: We created one BibTeX entry per source and replaced every hand-written citation with `\cite{key}` (Section I, line 19, line 33).

Changes in the revised paper: Section I, line 19, line 33

R1.2 [Addressed]

Reviewer comment: Los párrafos terminan con `\\`.

Response: All forced line breaks were removed; paragraphs are separated by a blank line.

Changes in the revised paper: Whole manuscript

Reviewer 2

Writing and structure

R2.1 [Addressed]

Reviewer comment: El artículo no formula una pregunta de investigación ni declara sus contribuciones.

Response: We added a research question and three contributions at the end of the Introduction (lines 23–31). The research question now reads:

“Can a single load cell under the cup holder detect cup placement and removal reliably enough, and fast enough, to release the robot automatically?”

Changes in the revised paper: Lines 23–31

Reviewer 3

Technical content

R3.1 [Addressed]

Reviewer comment: No se presenta la arquitectura del módulo ni la regla de detección.

Response: The architecture is shown in Fig. 1 (lines 40–44), and the detection rule with two thresholds is given in (1) (lines 49–51).

Changes in the revised paper: Fig. 1, Eq. (1), lines 40–51

R3.2 [Partially addressed]

Reviewer comment: No hay pruebas que respalden la confiabilidad de la detección.

Response: The test protocol was added in Table I, and the first 20 cycles are reported in lines 57–62. The complete campaign of 100 cycles will be reported in the next cut.

Changes in the revised paper: Table I, lines 57–62

Reviewer 4

D-Lab and state of the art

R4.1 [Addressed]

Reviewer comment: No se explica por qué RFID no es adecuado para RoboCoffee.

Response: Related Work now explains that RFID needs a tag on every disposable cup (lines 34–38).

Changes in the revised paper: Lines 34–38

R4.2 [Not addressed (justified)]

Reviewer comment: Implementen una segunda versión del módulo con cámara y comparen ambas.

Response: We agree that a camera could identify the cup independently of its weight. However, image processing and controlled lighting are outside the scope of this module. Instead, vision-based detection is discussed as future work in lines 66–69.

Changes in the revised paper: Lines 64–69